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Experiment No.	2

AIM:	**Aim:** To understand and implement Static and Dynamic Linking
CODE :	

1) **lib_mylib.c**

```
#include <stdio.h>

int reverseNumber(int n) {
    int rev = 0, rem;
    while (n != 0) {
        rem = n % 10;
        rev = rev * 10 + rem;
        n = n / 10;
    }
    return rev;
}

int sumOfDigits(int n) {
    int sum = 0, rem;
    while (n != 0) {
        rem = n % 10;
        sum += rem;
        n = n / 10;
    }
    return sum;
}
```

2) **lib_mylib.c**

```
#include <stdio.h>

int fibonacci(int n) {
    int a = 0, b = 1, c;
    if (n == 0)
        return a;
    for (int i = 2; i <= n; i++) {
        c = a + b;
        a = b;
        b = c;
    }
    return b;
}

int countDigits(int n) {
    int count = 0;
    while (n != 0) {
        count++;
        n = n / 10;
    }
    return count;
}
```

3) **lib_mylib.h**

```
int reverseNumber(int n);
int sumOfDigits(int n);
```

4) **lib_mylib.h**

```
int fibonacci(int n);
int countDigits(int n);
```

5) **driver.c**

```
#include <stdio.h>
#include "lib_mylib.h"
#include "lib_mylib1.h"

int main() {
    printf("Reverse of 1234 = %d\n", reverseNumber(1234));
    printf("Sum of digits of 567 = %d\n", sumOfDigits(567));
    printf("Fibonacci term at position 7 = %d\n", fibonacci(7));
    printf("Number of digits in 98765 = %d\n", countDigits(98765));

    return 0;
}
```

OUTPUT

Dynamic Linking

```
arsalaan@students:~/Desktop/Project$ gcc -c lib_mylib.c
arsalaan@students:~/Desktop/Project$ gcc -c lib_mylib1.c
arsalaan@students:~/Desktop/Project$ gcc -fPIC -c lib_mylib.c
arsalaan@students:~/Desktop/Project$ gcc -fPIC -c lib_mylib1.c
arsalaan@students:~/Desktop/Project$ gcc -shared -o libmylib.so lib_mylib.o lib_mylib1.o
arsalaan@students:~/Desktop/Project$ gcc -c driver.c
arsalaan@students:~/Desktop/Project$ gcc driver.o -L. -lmylib -o dynamic_program
arsalaan@students:~/Desktop/Project$ export LD_LIBRARY_PATH=.
arsalaan@students:~/Desktop/Project$ ./dynamic_program
Reverse of 1234 = 4321
Sum of digits of 567 = 18
Fibonacci term at position 7 = 13
Number of digits in 98765 = 5
arsalaan@students:~/Desktop/Project$
```

Static Linking

```
arsalaan@students:~/Desktop/Project$ gcc -c lib_mylib.c
arsalaan@students:~/Desktop/Project$ gcc -c lib_mylib1.c
arsalaan@students:~/Desktop/Project$ gcc -fPIC -c lib_mylib.c
arsalaan@students:~/Desktop/Project$ gcc -fPIC -c lib_mylib1.c
arsalaan@students:~/Desktop/Project$ gcc -shared -o libmylib.so lib_mylib.o lib_mylib1.o
arsalaan@students:~/Desktop/Project$ gcc -c driver.c
arsalaan@students:~/Desktop/Project$ gcc driver.o -L. -lmylib -o dynamic_program
arsalaan@students:~/Desktop/Project$ export LD_LIBRARY_PATH=.
arsalaan@students:~/Desktop/Project$ ./dynamic_program
Reverse of 1234 = 4321
Sum of digits of 567 = 18
Fibonacci term at position 7 = 13
Number of digits in 98765 = 5
arsalaan@students:~/Desktop/Project$
```

CONCLUSION:

In this program, we created two libraries with different functions and used them in a main program. We learned how to **link them statically and dynamically**, showing the difference between embedding library code into the executable (static) and loading it at runtime (dynamic). Dynamic linking reduces executable size and allows library code to be shared, while static linking makes the program independent of external files. This approach helps organize code, reuse functions, and manage large projects efficiently.